ACTIVE DIRECTORY
CERTIFICATE SERVICES
AD CS → EFS

Internal PKI Certificate Enrolment & Encrypting File System Demonstration

Lab: sahbk.com | CA: SAHKBAK.COM-CA | Platform: Windows Server 2022 + Windows 8.1

AD CS	EFS	PKI	ISO
Internal Certificate Authority Role	Encrypting File System	Public Key Infrastructure Built	27001 Controls Demonstrated

Date: 24 August 2026

SAHEED BAKARE

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1. Executive Summary

This document presents the complete technical implementation record for an Active Directory Certificate Services (AD CS) and Encrypting File System (EFS) project carried out in the Sahbak on-premises lab environment. The project establishes a functional internal Public Key Infrastructure (PKI) by deploying an Enterprise Root Certificate Authority, enrolling domain users for EFS-capable certificates, and demonstrating certificate-based file encryption and access control.

The lab environment uses Windows Server 2022 running Active Directory Domain Services (AD DS), DNS, and Active Directory Certificate Services within the sahbak.com domain. A domain-joined Windows client is used for certificate enrolment, EFS configuration, encryption testing, and access verification.

The project demonstrates the end-to-end relationship between identity, certificates, and encryption. Active Directory provides the user identity, AD CS issues a trusted certificate associated with that identity and its public/private key pair, and EFS uses the certificate to protect sensitive files. Access is then tested using an authorised user and a separate domain user who does not possess the required private key, demonstrating that successful domain authentication alone does not grant access to EFS-encrypted data.

This project demonstrates core PKI and IAM security concepts including certificate trust, public-key cryptography, identity binding, confidentiality, certificate enrolment, least privilege, and cryptographic access control.

2. Project Objectives

This project was designed to demonstrate a practical, end-to-end implementation of enterprise identity-linked cryptography using only built-in Windows Server and client components — requiring no third-party software.

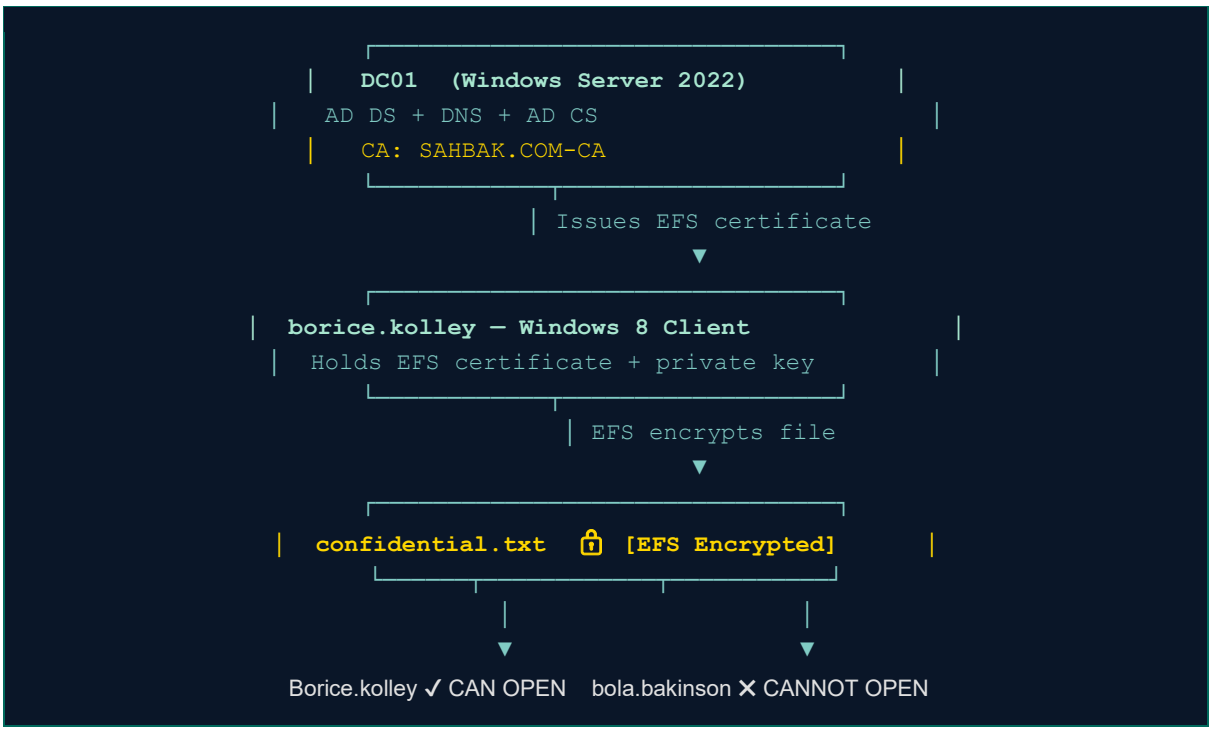
#	Objective	How the Lab Demonstrates It
O-1	**Deploy an internal Enterprise Root Certificate Authority using AD CS**	AD CS is installed on Windows Server 2022 and configured as an Enterprise Root CA for the sahbak.com domain, creating the internal certificate trust foundation for the lab.
O-2	**Issue an EFS-capable certificate to a domain user**	An authorised Sahbak domain user enrols for an EFS-capable certificate through Active Directory Certificate Services. The certificate is issued and trusted by the internal Sahbak Certificate Authority.
O-3	**Use the user certificate to encrypt a file with EFS**	The authorised user creates a confidential test file and enables Encrypt contents to secure data. EFS uses the user's certificate and associated cryptographic keys to protect the file contents.
O-4	**Demonstrate that another user cannot decrypt the file**	A second valid Sahbak domain user attempts to access the encrypted file but cannot decrypt its contents because they do not possess the authorised user's EFS private key.
O-5	**Show the relationship between AD identity, PKI, certificates, encryption, and access control**	The complete trust process is demonstrated: AD DS identity → AD CS certificate issuance → certificate/public-private key pair → EFS file encryption → authorised private-key-based decryption.

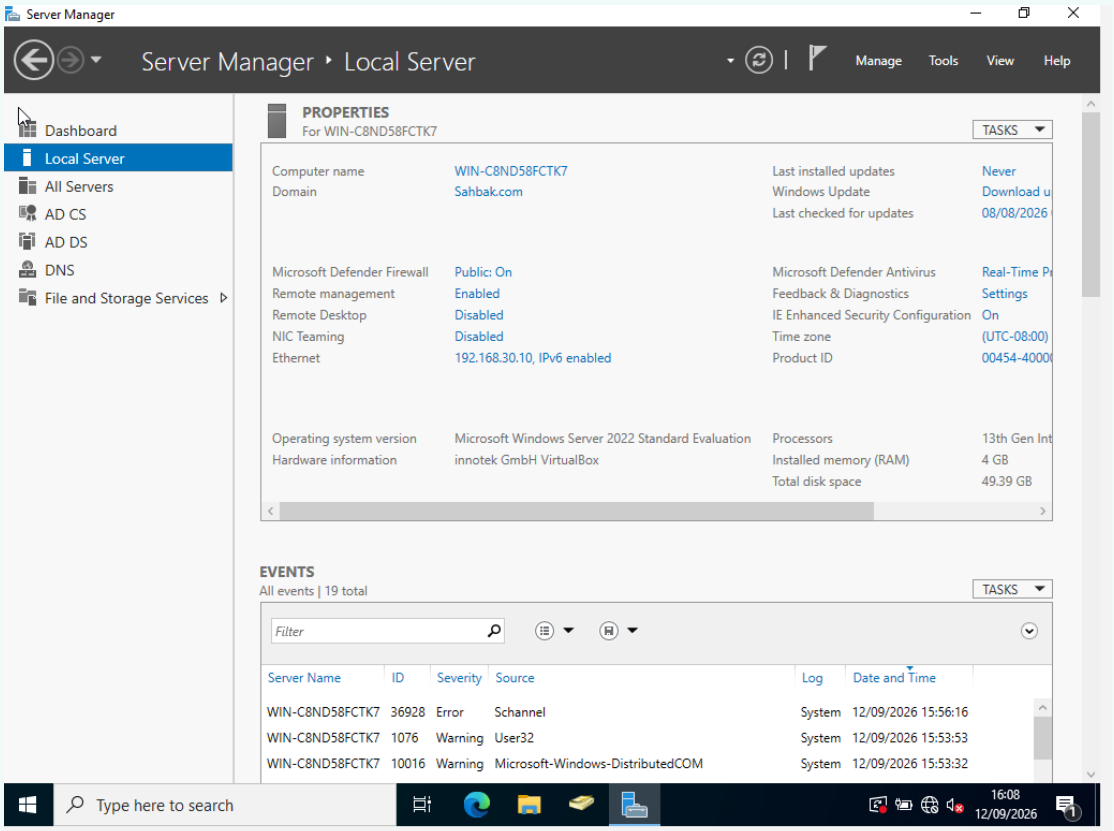
3. Lab Environment & Architecture

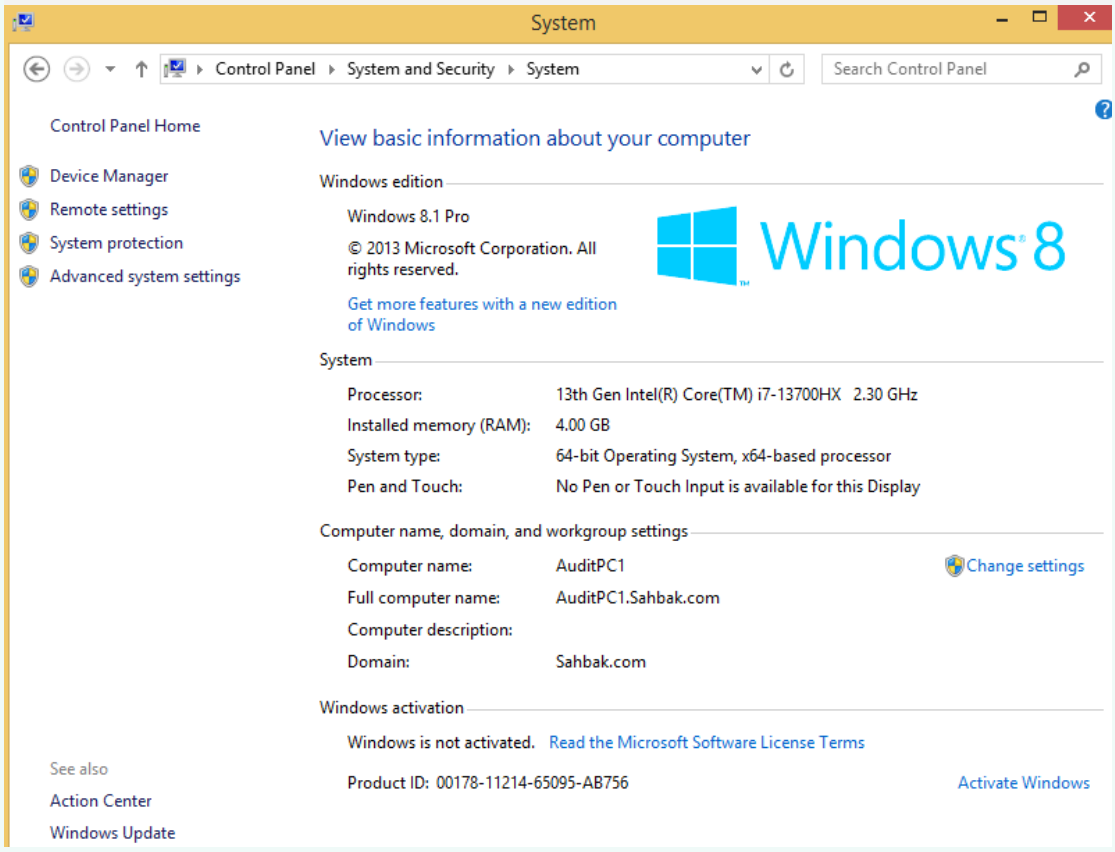
3.1 Lab Components

Component	Platform / Role	Name / Value	Purpose
Domain Controller	Windows Server 2022	`DC-01`	Hosts AD DS, DNS, and the AD CS Certificate Authority role
Active Directory Domain	`sahbak.com`	`sahbak.com`	Central identity and trust environment for the lab
Certificate Authority	AD CS Enterprise Root CA	`SAHBAK.COM-CA`	Issues and signs internal certificates; root of trust for the PKI
Client Workstation	Windows 8 (domain-joined)	`WIN8-CLIENT`	Certificate enrolment endpoint; EFS test machine
Test User 1 (Owner)	Domain User Account	`Borice.kolley`	Enrols for EFS certificate; creates and encrypts the test file
Test User 2 (Attacker)	Domain User Account	`Bola.Bakinson`	Attempts to access the encrypted file without the required private key

3.2 Architecture Diagram




Server Manager
Server Manager ‣ Local Server
Manage Tools View Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storage Services
PROPERTIES
For WIN-C8ND58FCTK7
TASKS
Computer name WIN-C8ND58FCTK7
Domain Sahbak.com
Last installed updates Never
Windows Update Download u
Last checked for updates 08/08/2026
Microsoft Defender Firewall Public: On
Remote management Enabled
Remote Desktop Disabled
NIC Teaming Disabled
Ethernet 192.168.30.10, IPv6 enabled
Microsoft Defender Antivirus Real-Time P
Feedback & Diagnostics Settings
IE Enhanced Security Configuration On
Time zone (UTC-08:00)
Product ID 00454-4000
Operating system version Microsoft Windows Server 2022 Standard Evaluation
Hardware information innotek GmbH VirtualBox
Processors 13th Gen Int
Installed memory (RAM) 4 GB
Total disk space 49.39 GB
EVENTS
All events | 19 total
TASKS
Filter
Server Name ID Severity Source Log Date and Time
WIN-C8ND58FCTK7 36928 Error Schannel System 12/09/2026 15:56:16
WIN-C8ND58FCTK7 1076 Warning User32 System 12/09/2026 15:53:53
WIN-C8ND58FCTK7 10016 Warning Microsoft-Windows-DistributedCOM System 12/09/2026 15:53:32
Type here to search
16:08
12/09/2026


System
Control Panel ‣ System and Security ‣ System
Search Control Panel
Control Panel Home
Device Manager
Remote settings
System protection
Advanced system settings
View basic information about your computer
Windows edition
Windows 8.1 Pro
© 2013 Microsoft Corporation. All rights reserved.
Get more features with a new edition of Windows
Windows 8
System
Processor: 13th Gen Intel(R) Core(TM) i7-13700HX 2.30 GHz
Installed memory (RAM): 4.00 GB
System type: 64-bit Operating System, x64-based processor
Pen and Touch: No Pen or Touch Input is available for this Display
Computer name, domain, and workgroup settings
Computer name: AuditPC1
Change settings
Full computer name: AuditPC1.Sahbak.com
Computer description:
Domain: Sahbak.com
Windows activation
Windows is not activated. Read the Microsoft Software License Terms
Product ID: 00178-11214-65095-AB756
Activate Windows
See also
Action Center
Windows Update

4. Conceptual Foundation — PKI, AD CS, and EFS

4.1 What is a Public Key Infrastructure (PKI)?

A Public Key Infrastructure (PKI) is the framework of policies, procedures, hardware, software, and people needed to create, manage, distribute, use, store, and revoke digital certificates. At its core, PKI solves a fundamental problem in cryptography: how do you know that a public key actually belongs to the person it claims to belong to?

PKI addresses this by introducing a trusted third party — the Certificate Authority (CA) — that digitally signs certificates. A certificate is a document binding a public key to an identity (e.g. borice.kolley@sahbak.com). Because the CA's signature can be verified by anyone who trusts the CA, a certificate provides proof of identity with cryptographic strength.

PKI Key Concepts

Certificate Authority (CA): The trusted entity that issues and signs certificates. In this lab: SAHBAK.COM-CA

Certificate: A digitally signed document containing a public key, the identity it belongs to, its validity period, and the CA's signature

Public Key: The part of the key pair that can be freely shared — used to encrypt data or verify signatures

Private Key: The secret part of the key pair — held by the certificate owner and used to recover the encryption key required to decrypt protected data

Trust Chain: A path of cryptographic trust from a root CA, through any intermediate CAs, down to the end-entity certificate. Trust in the root CA means trust in everything it signs

4.2 What is Active Directory Certificate Services (AD CS)?

Active Directory Certificate Services (AD CS) is a Windows Server role that provides the services needed to create and manage a PKI tied to your Active Directory environment. An Enterprise CA installed as part of AD CS is tightly integrated with AD DS — it reads domain user accounts and groups from Active Directory to determine who is authorised to enrol for which certificate templates, and it publishes issued certificates back to Active Directory.

AD CS Capability	What It Means in This Lab
Enterprise Root CA	SAHBAK.COM-CA is the top-level trust anchor — it signs all certificates issued in this domain
Certificate Templates	Pre-configured certificate profiles (e.g. the EFS template) define what a certificate can be used for and who can enrol for it
Active Directory Integration	The CA uses AD to verify borice.kolley@sahbak.com is a valid domain user before issuing a certificate — AD identity is the basis of certificate trust
Automatic Enrolment	Through Group Policy, AD CS can automatically provision certificates to eligible users or computers based on certificate template permissions and auto-enrolment configuration.
Certificate Revocation	The CA can revoke certificates that are compromised or no longer trusted. Revocation prevents continued trust in the certificate for supported validation scenarios, while previously encrypted EFS data may still require separate key-management or recovery controls.

4.3 What is the Encrypting File System (EFS)?

The Encrypting File System (EFS) is a feature of the Windows NTFS file system that provides file-level encryption built directly into the operating system. When a file is encrypted with EFS, Windows transparently encrypts and decrypts it as the authorised user reads and writes it — but any other user who attempts to access the file receives an 'Access is denied' error, even if they have NTFS read permissions.

Crucially, EFS is not the same as NTFS permissions. NTFS permissions control who can see a file in the directory listing and access its metadata — they are enforced by the Windows security subsystem. EFS encryption protects the actual content of the file using cryptography — it is enforced by the mathematics of asymmetric encryption and cannot be bypassed by administrators with NTFS access unless they also hold the correct decryption key.

EFS vs NTFS Permissions — A Critical Distinction

NTFS Permissions (traditional access control):

- Controlled by ACLs (Access Control Lists) in the file system
- Enforced by Windows security subsystem (kernel)
- Can be overridden by domain admins with 'Take Ownership' right
- Protects access at the file-system level

EFS Encryption (cryptographic access control):

- Controlled by possession of the private key
- Enforced by asymmetric cryptography (mathematics, not policy)
- Cannot be bypassed by admins without the private key (unless a DRA is configured)
- Protects the actual file contents — even if the file is copied to another system

4.4 How AD CS and EFS Work Together

→	Component	Role in the Chain
1	**Active Directory (AD DS)**	Provides the user identity — borice.kolley is a domain account in sahbak.com. Active Directory is the identity foundation.
2	**AD CS / SAHBAK.COM-CA**	Issues an EFS certificate to borice.kolley. The certificate contains the user's public key and is digitally signed by the CA, establishing trust.
3	**Certificate**	The issued certificate binds borice.kolley’s public key to their Active Directory identity. The corresponding private key is stored in the user’s profile on the client.
4	**EFS Encryption**	When borice.kolley encrypts a file, EFS generates a random symmetric File Encryption Key (FEK). EFS then encrypts the FEK using borice.kolley public key (from the certificate) and stores the encrypted FEK in the file's metadata.
5	**EFS Decryption (borice.kolley)**	When borice.kolley opens the file, EFS uses borice.kolley private key to decrypt the FEK, then uses the FEK to decrypt the file content. This happens transparently.
6	**Denial (bola.bakinson)**	When bola.bakinson attempts to open the file, EFS cannot recover the encrypted FEK because bola.bakinson does not possess the private key corresponding to borice.kolley’s EFS certificate. Access is denied.

5. Implementation — Step-by-Step

The following ten steps document the complete implementation in the order it was performed. Each step includes the exact navigation path, configuration values used, and screenshot evidence placeholders at every key point. The document follows the authoritative sysadmin convention: record every decision and its rationale, not just the actions.

01 **Install the AD CS Role (Server Manager)** — Windows Server 2022 — Server Manager

5.1 Step 1 — Install the AD CS Role

The Active Directory Certificate Services role was installed on IT-Server using Server Manager's Add Roles and Features Wizard. Only the Certification Authority role service was selected. This server will later be configured as the Root Certificate Authority named SAHBAK.COM-CA. Other AD CS role services—including NDES, CES, CEP and Web Enrollment—were not installed because they were outside the scope of this foundational lab.

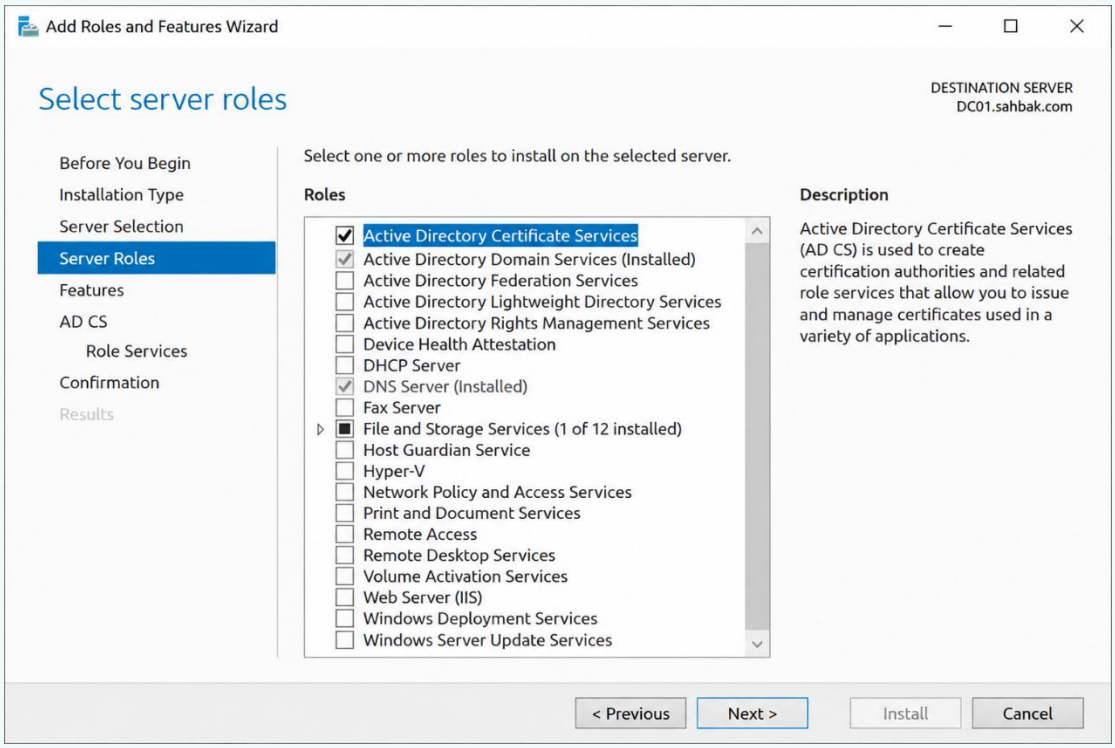
Navigation Path

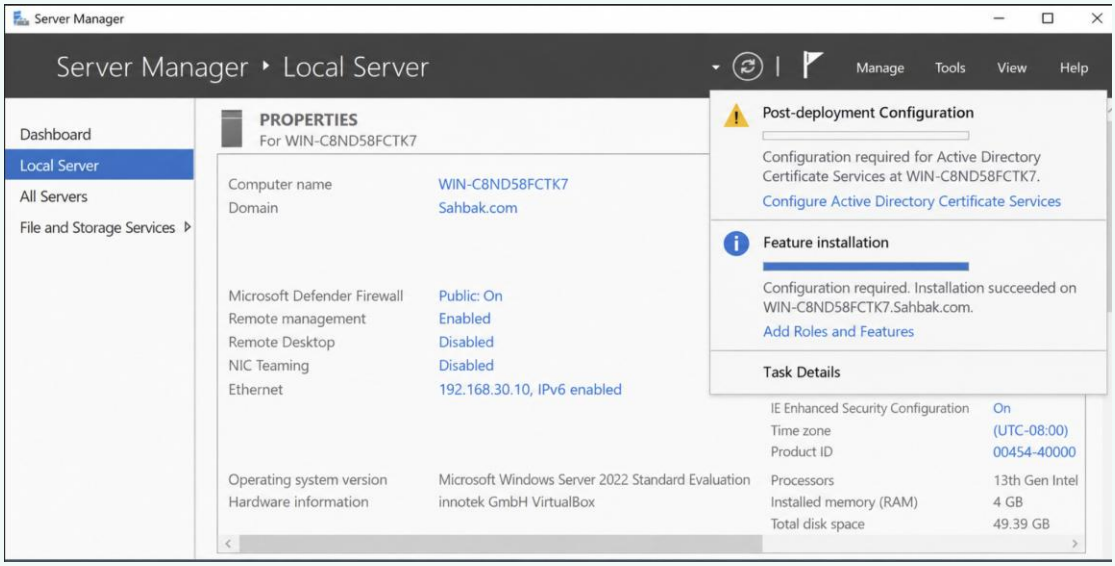
- Server Manager > Dashboard > Add roles and features
- Installation Type: Role-based or feature-based installation
- Server Selection: IT-Server (the local server)
- Server Roles: Active Directory Certificate Services — checked
- Role Services: Certification Authority — checked (only this one)
- Confirmation page: checked Restart the destination server automatically if required
- Clicked Install — waited for installation to complete

▶ PowerShell — AD CS Role Installation (Equivalent Command)

```
# PowerShell equivalent — install AD CS role
Install-WindowsFeature -Name AD-Certificate -IncludeManagementTools

# Verify the role was installed
Get-WindowsFeature -Name AD-Certificate
[X] Active Directory Certificate Services   AD-Certificate
[X]   Certification Authority               ADCS-Cert-Authority
```





Sysadmin Note — Why Only the Certification Authority Role Service?

The AD CS role includes several additional role services: Certification Authority Web Enrollment, Network Device Enrollment Service (NDES), Online Responder, Certificate Enrollment Web Service (CES) and Certificate Enrollment Policy Web Service (CEP). For this foundational lab, which demonstrates Active Directory-integrated certificate enrollment through Microsoft Management Console (MMC) and Group Policy, only the Certification Authority role service is required. Excluding unnecessary role services reduces the attack surface and keeps the implementation aligned with the project objectives.

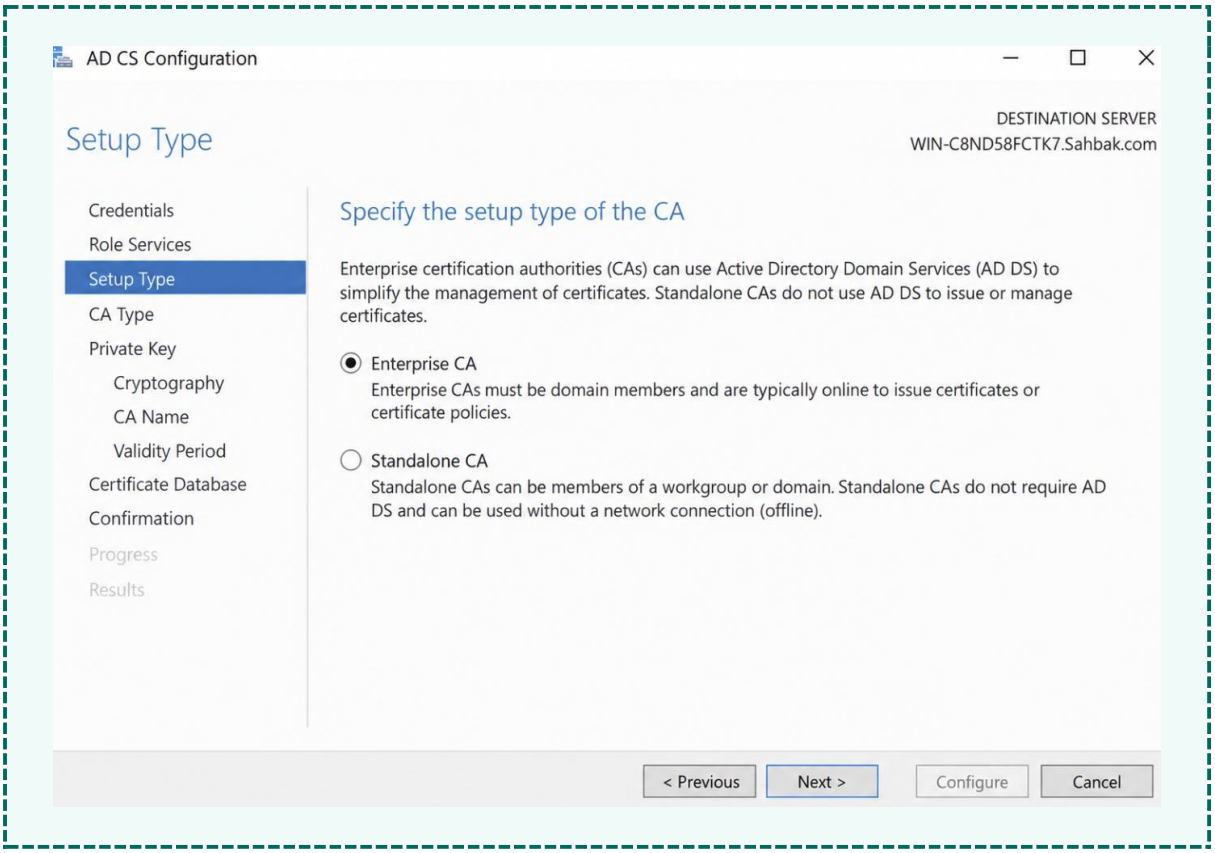
02 **Configure the Enterprise Root CA** AD CS Configuration Wizard

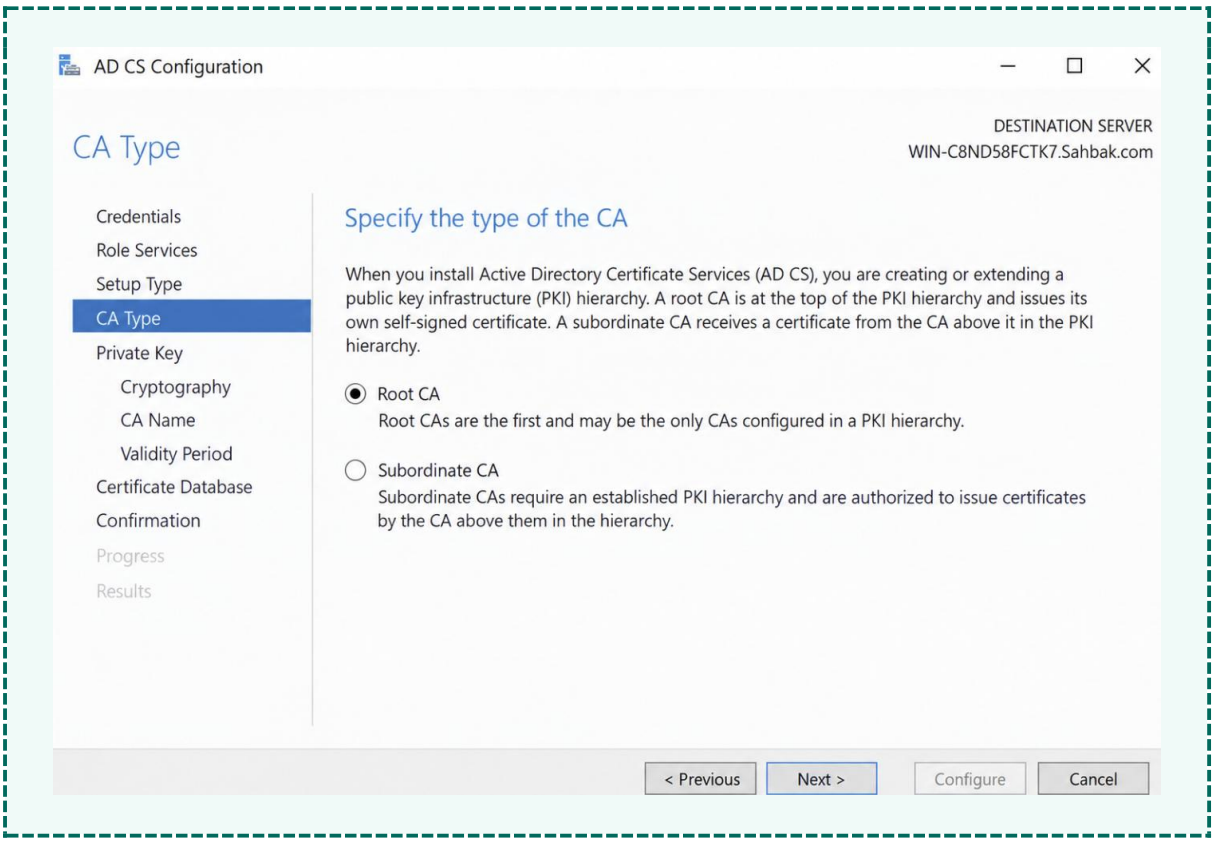
5.2 Step 2 — Configure the Enterprise Root CA

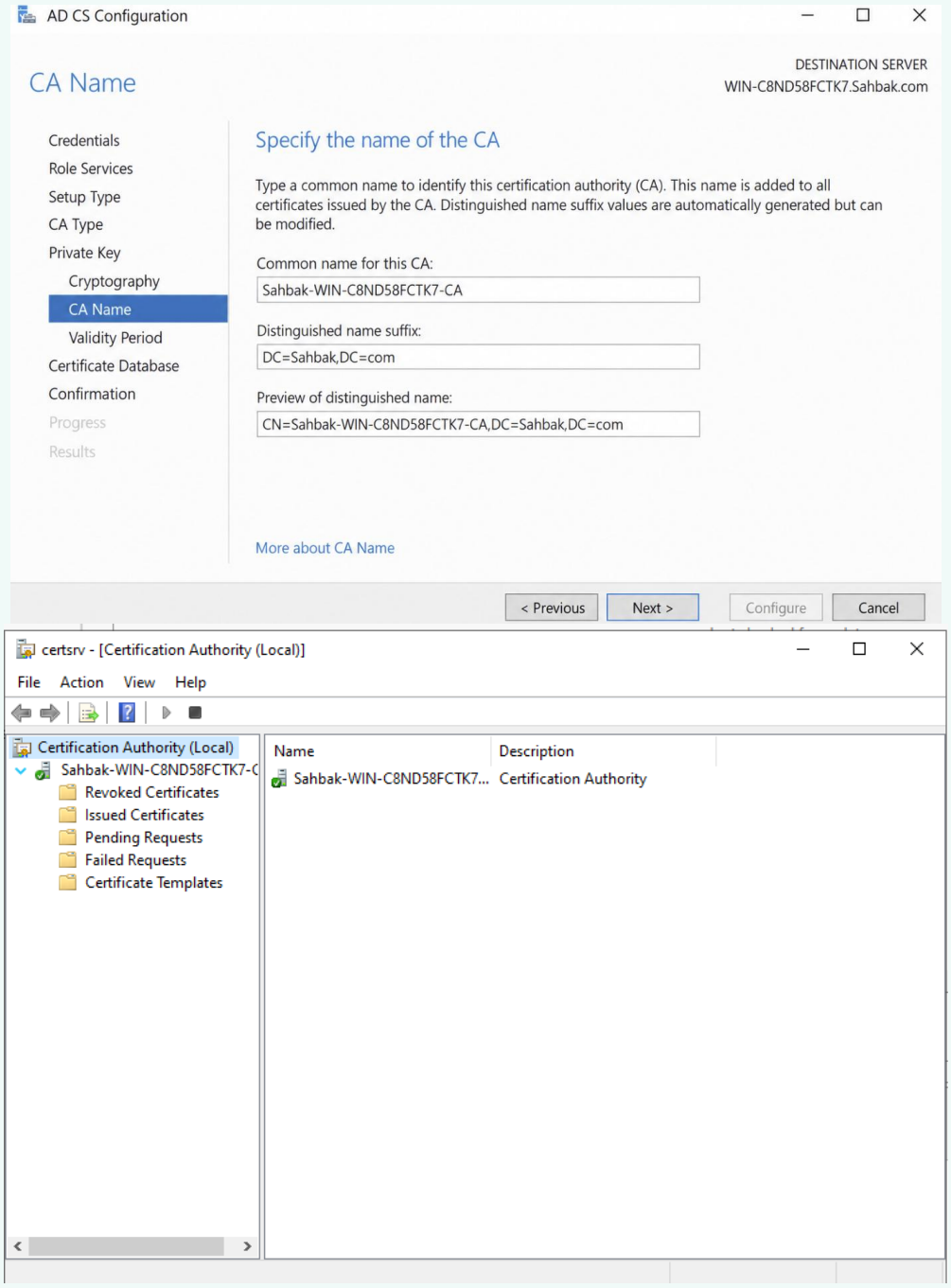
After the AD CS role was installed, the post-deployment configuration wizard was launched to configure the Certification Authority. An Enterprise Root CA was selected because this is a new Active Directory-integrated PKI hierarchy. The Enterprise CA uses certificate templates and Active Directory permissions to manage certificate enrollment. A new private key was generated for the CA.

Configuration Values Used

Configuration Setting	Value Selected / Rationale
Credentials	Domain Administrator account — used to configure the Enterprise CA
Role Services to Configure	Certification Authority
Setup Type	Enterprise CA — integrates with Active Directory, certificate templates and domain-based enrollment permissions
CA Type	Root CA — top-level CA in a new PKI hierarchy; no parent CA to sign this CA's certificate
Private Key	Create a new private key — generates a new cryptographic key pair for the Certification Authority
Cryptographic Provider	RSA#Microsoft Software Key Storage Provider
Key Length	2048 bits — selected for this lab; production key length should follow the organisation’s cryptographic policy
Hash Algorithm	SHA-256 — selected for certificate signing; legacy MD5 and SHA-1 algorithms were avoided
CA Common Name	SAHBAK.COM-ROOT-CA — reflects the domain/organisation context
Distinguished Name Suffix	DC=Sahbak,DC=com
Validity Period	5 years — selected for the purposes and expected lifecycle of this lab
Certificate Database	Default path: C:\Windows\system32\CertLog


AD CS Configuration
Setup Type
DESTINATION SERVER
WIN-C8ND58FCTK7.Sahbak.com
Credentials
Role Services
Setup Type
CA Type
Private Key
Cryptography
CA Name
Validity Period
Certificate Database
Confirmation
Progress
Results
Specify the setup type of the CA
Enterprise certification authorities (CAs) can use Active Directory Domain Services (AD DS) to simplify the management of certificates. Standalone CAs do not use AD DS to issue or manage certificates.
Enterprise CA
Enterprise CAs must be domain members and are typically online to issue certificates or certificate policies.
Standalone CA
Standalone CAs can be members of a workgroup or domain. Standalone CAs do not require AD DS and can be used without a network connection (offline).
< Previous
Next >
Configure
Cancel


AD CS Configuration
CA Type
DESTINATION SERVER
WIN-C8ND58FCTK7.Sahbak.com
Credentials
Role Services
Setup Type
CA Type
Private Key
Cryptography
CA Name
Validity Period
Certificate Database
Confirmation
Progress
Results
Specify the type of the CA
When you install Active Directory Certificate Services (AD CS), you are creating or extending a public key infrastructure (PKI) hierarchy. A root CA is at the top of the PKI hierarchy and issues its own self-signed certificate. A subordinate CA receives a certificate from the CA above it in the PKI hierarchy.
Root CA
Root CAs are the first and may be the only CAs configured in a PKI hierarchy.
Subordinate CA
Subordinate CAs require an established PKI hierarchy and are authorized to issue certificates by the CA above them in the hierarchy.
< Previous
Next >
Configure
Cancel



Enterprise CA vs Standalone CA — Why Enterprise Was Chosen

An Enterprise CA is deeply integrated with Active Directory:

- Uses AD to store certificate templates, making them available for enrollment to domain users
- Automatically publishes issued certificates to the AD user object
- Supports automatic certificate enrollment through Group Policy
- Appears as a trusted CA automatically in domain-joined machines' certificate stores

A Standalone CA has none of these AD integrations — it is appropriate for workgroup environments or issuing certificates to non-domain entities. For this lab's purpose of demonstrating identity-linked certificate issuance via the healthcare.local domain, an Enterprise CA is the only appropriate choice.

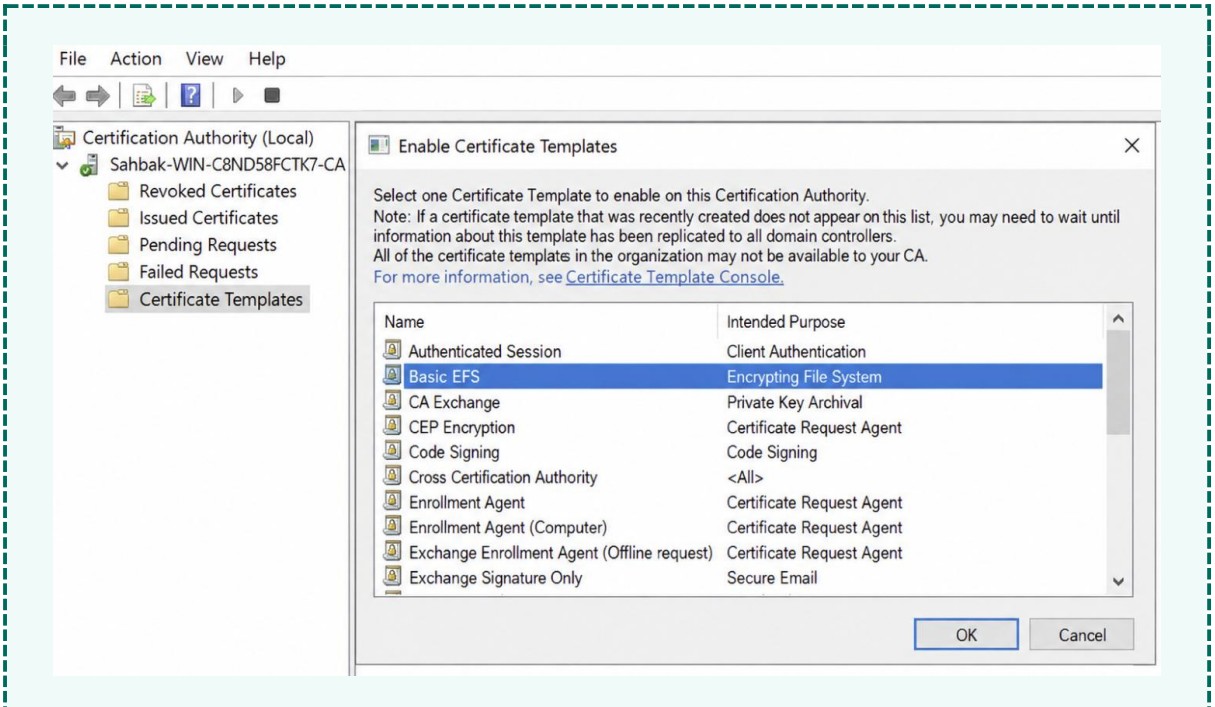
03 **Enable the EFS Certificate Template** | Certification Authority MMC (certsrv.msc)

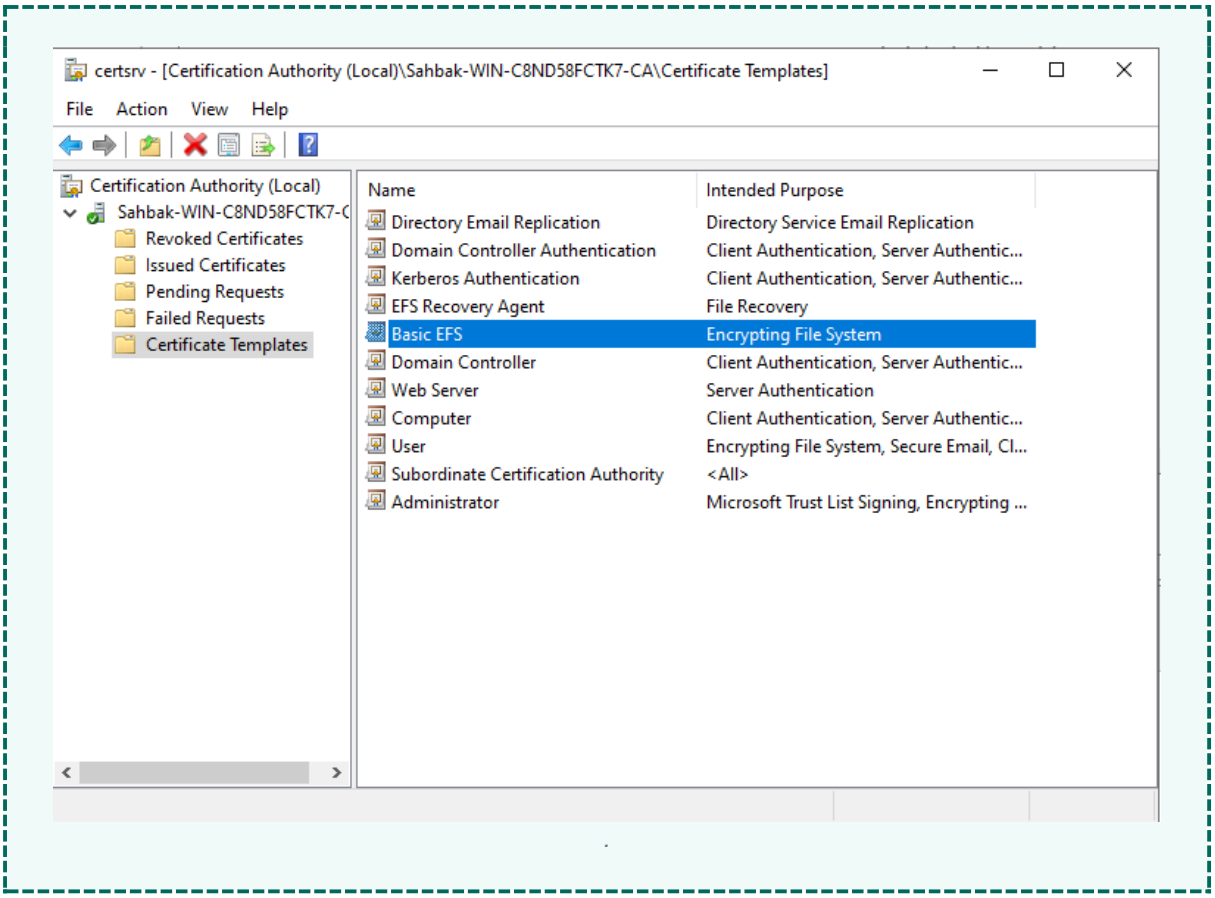
5.3 Step 3 — Enable the EFS Certificate Template

Certificate templates define the purpose of a certificate, who is permitted to enrol for it, its validity period and its cryptographic settings. Windows Server includes several built-in certificate templates. For this project, the Basic EFS template was enabled within the SAHBAK-ROOT-CA domain to allow authorised users to request certificates for Encrypting File System (EFS). These certificates enable users to encrypt and decrypt files securely within the domain environment.

Enabling the Template in Certification Authority MMC

1. On IT-Server, opened certsrv.msc (Start > Run > certsrv.msc)
2. Expanded SAHBAK-ROOT-CA in the left panel
3. Right-clicked Certificate Templates > New > Certificate Template to Issue
4. The Enable Certificate Templates dialog appeared — listing all available templates
5. Selected Basic EFS from the list
6. Clicked OK — the template is now available for enrollment by domain users
7. Verified the template appeared under Certificate Templates in the CA console




certsrv - [Certification Authority (Local)\Sahbak-WIN-C8ND58FCTK7-CA\Certificate Templates]
File
Action
View
Help
Certification Authority (Local)
Sahbak-WIN-C8ND58FCTK7-C
Revoked Certificates
Issued Certificates
Pending Requests
Failed Requests
Certificate Templates
Name
Intended Purpose
Directory Email Replication
Directory Service Email Replication
Domain Controller Authentication
Client Authentication, Server Authentic...
Kerberos Authentication
Client Authentication, Server Authentic...
EFS Recovery Agent
File Recovery
Basic EFS
Encrypting File System
Domain Controller
Client Authentication, Server Authentic...
Web Server
Server Authentication
Computer
Client Authentication, Server Authentic...
User
Encrypting File System, Secure Email, Cl...
Subordinate Certification Authority
<All>
Administrator
Microsoft Trust List Signing, Encrypting ...

04 **Create Domain Users — ali.test & other.test**

Active Directory Users and Computers (ADUC)

5.4 Step 4 — Select Existing EFS Test Users

Two existing SAHBAK.COM domain accounts were selected for the EFS demonstration. borice.koley@sahbak.com was assigned as the certificate holder and file owner. kola.abiola@sahbak.com was selected as the second user to test access restrictions and confirm that another user could not decrypt the EFS-protected file without the appropriate certificate and private key.

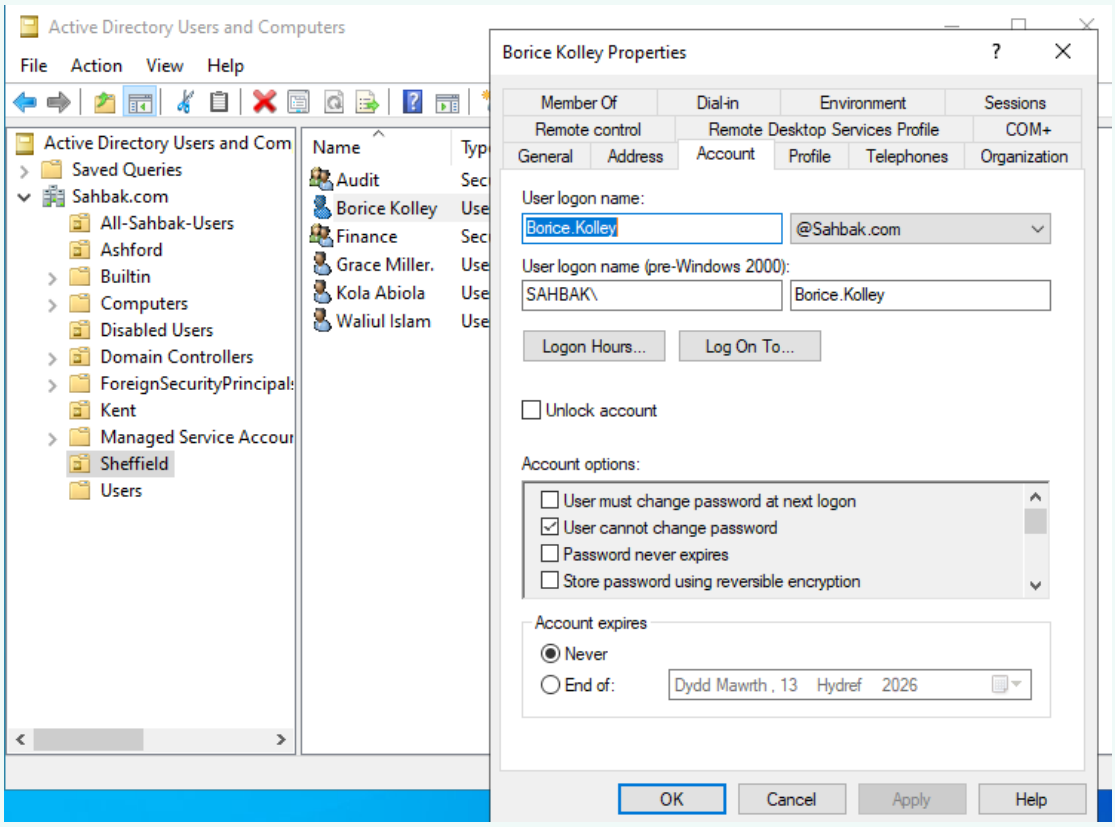
Both accounts were originally provisioned during the on-premises IAM implementation and are members of the **Finance** security group within the **Sheffield** organisational unit. Reusing existing accounts maintains consistency between the IAM and AD CS/EFS projects.

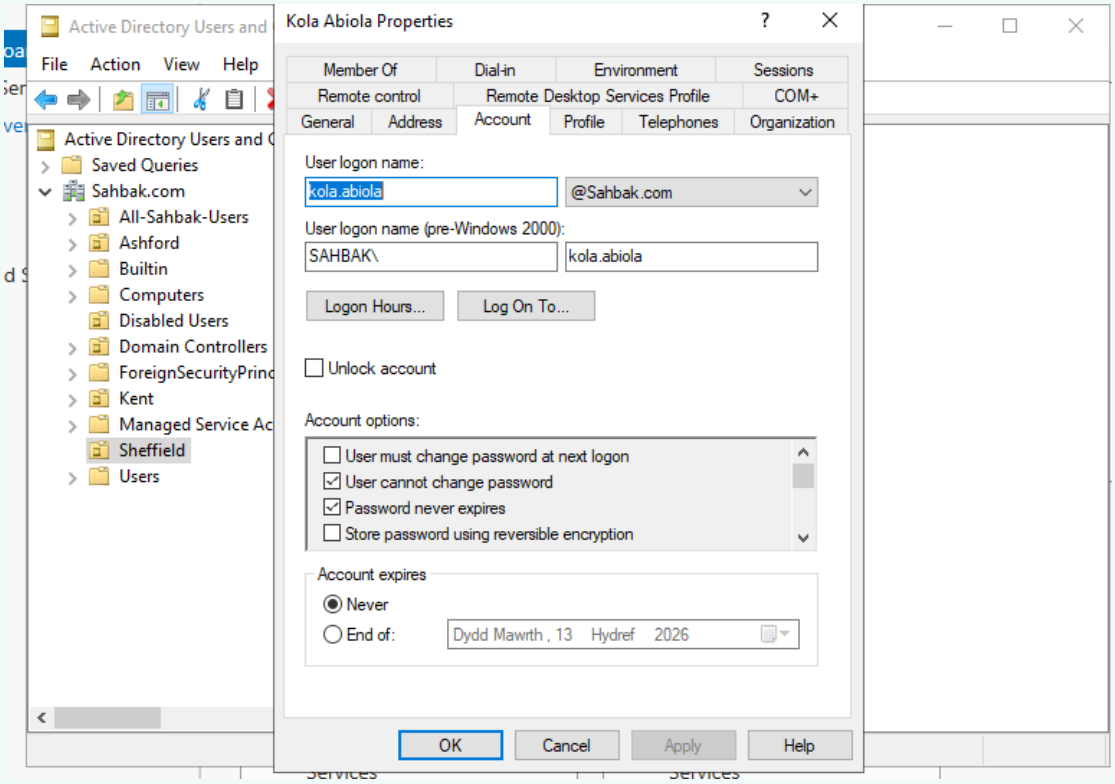
Verifying the EFS File Owner

8. On the domain server, opened Active Directory Users and Computers by running dsa.msc.
9. Navigated to SAHBAK.COM > Sheffield.
10. Located and opened the properties of the Finance user account.
11. Verified the account details for borice.koley@sahbak.com.
12. Assigned this account as the certificate holder and owner of the EFS test file.

Verifying the Access-Test USer

13. Located the second Finance account in the Sheffield organisational unit.
14. Verified the account details for kola.abiola@sahbak.com.
15. Assigned this account as the secondary test user.
16. Confirmed that both accounts were enabled and available for the EFS access-control test.


Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
Sahbak.com
All-Sahbak-Users
Ashford
Builtin
Computers
Disabled Users
Domain Controllers
ForeignSecurityPrincipal
Kent
Managed Service Accour
Sheffield
Users
Name
Audit
Borice Kolley
Finance
Grace Miller.
Kola Abiola
Waliul Islam
Borice Kolley Properties
Member Of
Dial-in
Environment
Sessions
Remote control
Remote Desktop Services Profile
COM+
General
Address
Account
Profile
Telephones
Organization
User logon name:
Borice.Kolley
@Sahbak.com
User logon name (pre-Windows 2000):
SAHBAK\
Borice.Kolley
Logon Hours...
Log On To...
Unlock account
Account options:
User must change password at next logon
User cannot change password
Password never expires
Store password using reversible encryption
Account expires
Never
End of:
Dydd Mawrth , 13 Hydref 2026
OK
Cancel
Apply
Help

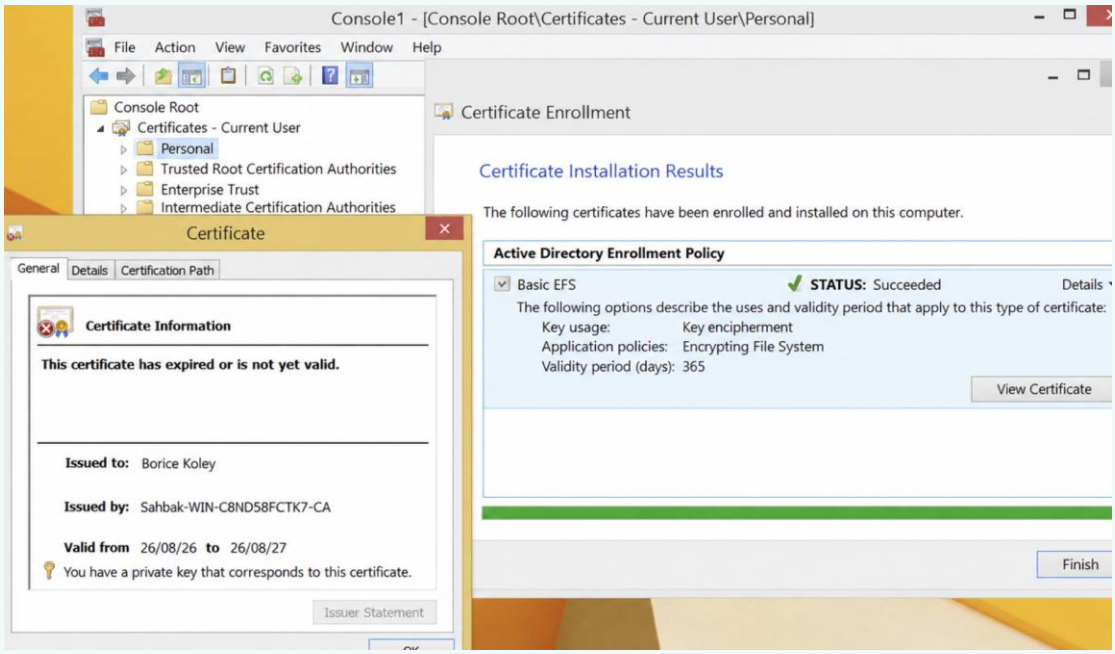

Active Directory Users and
File Action View Help
Active Directory Users and
Saved Queries
Sahbak.com
All-Sahbak-Users
Ashford
Builtin
Computers
Disabled Users
Domain Controllers
ForeignSecurityPrinc
Kent
Managed Service Ac
Sheffield
Users
Kola Abiola Properties
Member Of
Dial-in
Environment
Sessions
Remote control
Remote Desktop Services Profile
COM+
General
Address
Account
Profile
Telephones
Organization
User logon name:
kola.abiola
@Sahbak.com
User logon name (pre-Windows 2000):
SAHBAK\
kola.abiola
Logon Hours...
Log On To...
Unlock account
Account options:
User must change password at next logon
User cannot change password
Password never expires
Store password using reversible encryption
Account expires
Never
End of:
Dydd Mawrth , 13 Hydref 2026
OK
Cancel
Apply
Help

05 **Enrol Borice Kolley for an EFS Certificate** | **Certificates MMC (certmgr.msc) — Windows 8 Client**

5.5 Step 5 — Enrol Borice Kolley for an EFS Certificate

With the Certification Authority configured and the Basic EFS certificate template enabled, Borice Kolley signed in to a domain-joined client computer and submitted a certificate-enrolment request. The Certificates MMC snap-in was used to request the certificate through Active Directory Enrollment Policy. This demonstrated how an authorised SAHBAK.COM domain user can request an EFS certificate from an Enterprise Certification Authority.

17. On the domain-joined client, signed in as SAHBAK\Borice.Kolley.
18. Pressed Windows + R, entered certmgr.msc and selected Enter to open the Current User Certificates console
19. In the left pane, navigated to Certificates – Current User > Personal
20. Right-clicked Personal and selected All Tasks > Request New Certificate
21. The Certificate Enrollment wizard opened
22. Selected Active Directory Enrollment Policy and clicked Next
23. Reviewed the available certificate templates and selected Basic EFS.
24. Clicked Enroll. The Certification Authority processed the request according to the template's enrollment permissions and issued the certificate
25. Verified that the enrollment status displayed Succeeded and that the EFS certificate was installed in Borice Kolley's Personal certificate store



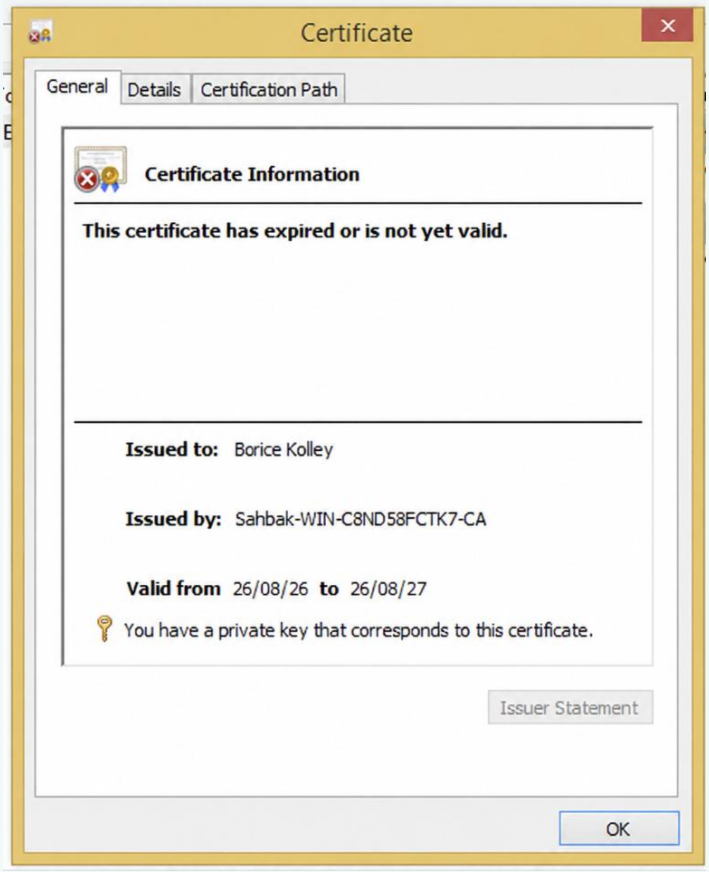
06 **Verify the Issued Certificate** | Certificates MMC — Current User → Personal → Certificates

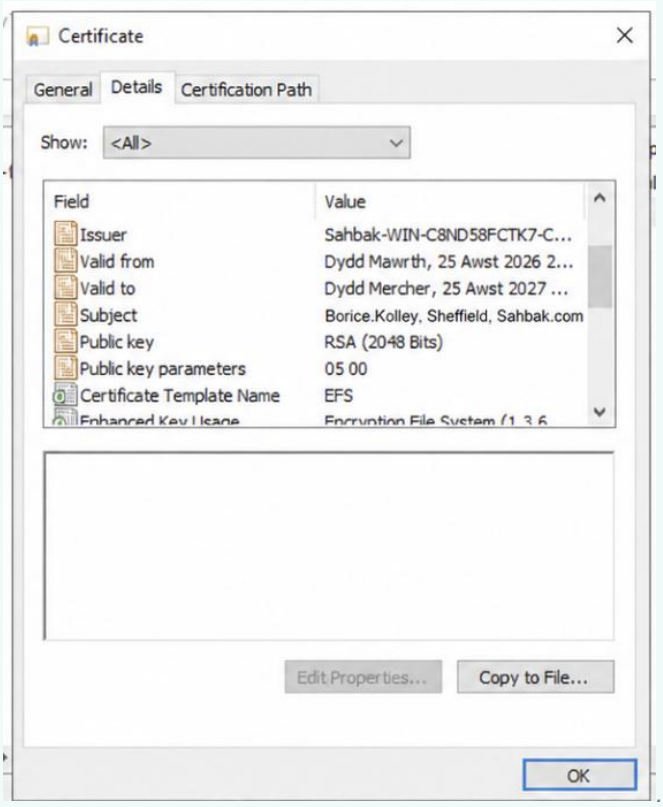
5.6 Step 6 — Verify the Issued Certificate

Following successful enrollment, the EFS certificate was inspected in the Current User Personal certificate store. The certificate was issued to Borice Kolley by the SAHBAK.COM-ROOT-CA. The certificate's subject, issuer, validity period, intended purpose and public-key information were reviewed before testing EFS encryption.

26. In certmgr.msc, navigated to Certificates — Current User > Personal > Certificates
27. The newly issued EFS certificate appeared in the right panel — issued to ali.test
28. Double-clicked the certificate to open the Certificate details dialog
29. Verified the following on the General tab:

Certificate Field	Expected / Confirmed Value
Issued to	Borice Kolley (borice.kolley@sahbak.com)
Issued by	SAHBAK-ROOT-CA
Valid from	26/08/2026 — confirms the certificate is currently valid
Valid to	[26/08/2027 — typically 1-2 years from issuance for Basic EFS template]
Intended purposes	Encrypting File System — confirms this certificate is for EFS use
Public key	RSA (2048 bits) — matches the template's cryptographic requirements
Subject	CN=Borice Kolley, OU=Sheffield, DC=Sahbak, DC=com
Issuer	CN=Sahbak-Root-CA, DC=Sahbak, DC=com


Certificate
General
Details
Certification Path
Certificate Information
This certificate has expired or is not yet valid.
Issued to: Borice Kolley
Issued by: Sahbak-WIN-C8ND58FCTK7-CA
Valid from 26/08/26 to 26/08/27
You have a private key that corresponds to this certificate.
Issuer Statement
OK


Certificate
General
Details
Certification Path
Show: <All>
Field
Value
Issuer
Sahbak-WIN-C8ND58FCTK7-C...
Valid from
Dydd Mawrth, 25 Awst 2026 2...
Valid to
Dydd Mercher, 25 Awst 2027 ...
Subject
Borice.Kolley, Sheffield, Sahbak.com
Public key
RSA (2048 Bits)
Public key parameters
05 00
Certificate Template Name
EFS
Edit Properties...
Copy to File...
OK

07 **Create the Confidential Test File** | Windows 8 Client — File Explorer / Notepad

5.7 Step 7 — Create the Confidential Test File

A test directory and file were created to serve as the encryption subject. In a real healthcare environment, this file would represent a confidential patient record, financial data, or clinical research data that must be protected at the file-system level against unauthorised access — even by other staff with access to the same machine or file share.

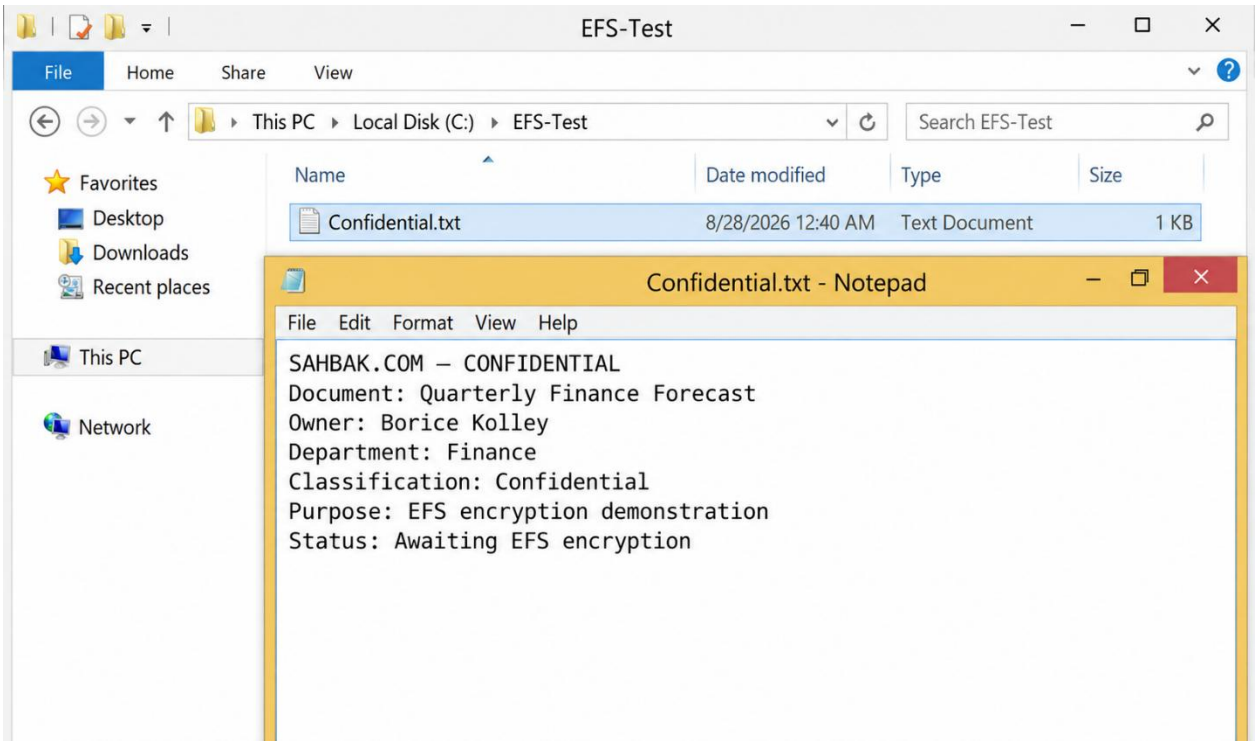
30. On the domain-joined client, signed in as SAHBAK\Borice.Kolley
31. Navigated to C:\ and created a new folder named EFS-Test
32. Opened Notepad and typed sample confidential content (e.g. patient name, date, notes)
33. Saved the file as C:\EFS-Test\confidential.txt
34. Closed Notepad

▶ **Create the Test File — PowerShell Equivalent**

```
# PowerShell equivalent — create test directory and file
New-Item -ItemType Directory -Path C:\EFS-Test
New-Item -ItemType File -Path C:\EFS-Test\confidential.txt

# Add non-sensitive sample content
Set-Content -Path C:\EFS-Test\confidential.txt -Value @'
FINANCIAL - CONFIDENTIAL
Owner: Borice Kolley
Department: Finance
Classification: Confidential
Purpose: EFS encryption demonstration Status: Awaiting EFS encryption
'@ | Set-Content -Path $TestFile -Encoding UTF8

# Verify file was created
Get-Item C:\EFS-Test\confidential.txt
```

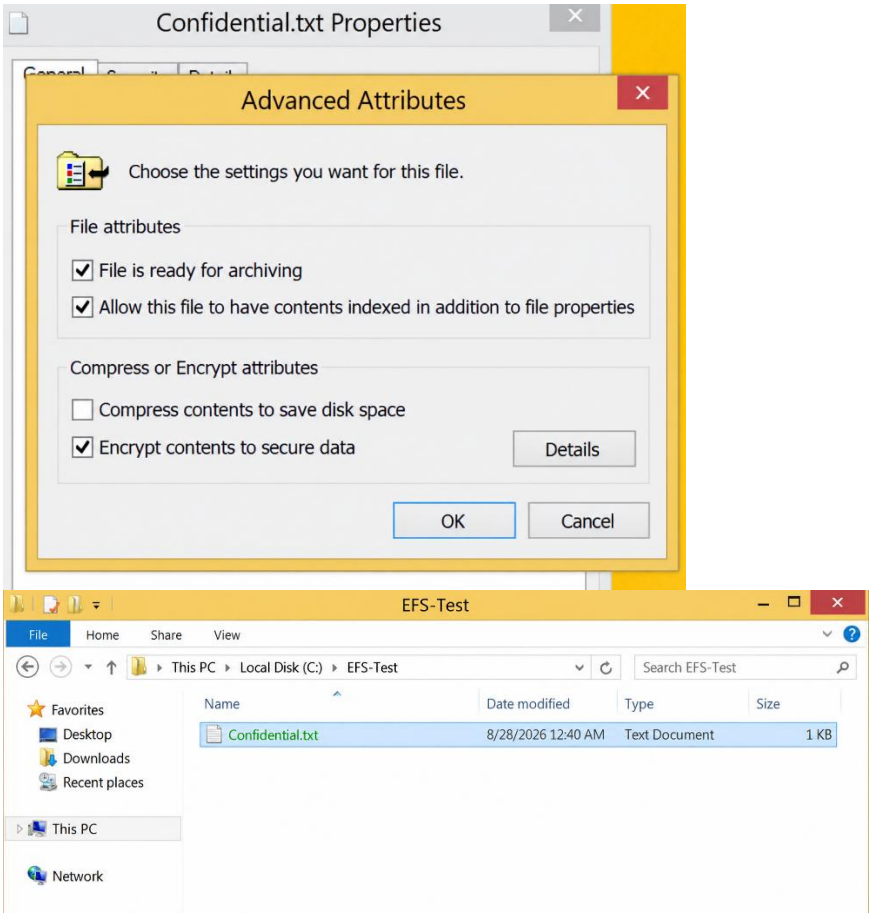

EFS-Test
File
Home
Share
View
This PC
Local Disk (C:)
EFS-Test
Search EFS-Test
Favorites
Desktop
Downloads
Recent places
This PC
Network
Name
Date modified
Type
Size
Confidential.txt
8/28/2026 12:40 AM
Text Document
1 KB
Confidential.txt - Notepad
File Edit Format View Help
SAHBAK.COM — CONFIDENTIAL
Document: Quarterly Finance Forecast
Owner: Borice Kolley
Department: Finance
Classification: Confidential
Purpose: EFS encryption demonstration
Status: Awaiting EFS encryption

08 **Encrypt the File with EFS** | File Properties → Advanced Attributes

5.8 Step 8 — Encrypt the File with EFS

The confidential test file was encrypted using Encrypting File System (EFS) through the Windows file-properties dialog. Windows encrypts the file contents using a randomly generated File Encryption Key (FEK). The FEK is then protected using the public key from Borice Kolley's enrolled EFS certificate. Borice Kolley's corresponding private key—or an authorised Data Recovery Agent, if configured—is required to recover the FEK and decrypt the file.

35. Signed in to the domain-joined client as SAHBAK\Borice.Kolley
36. Right-clicked C:\EFS-Test\Confidential.txt and selected Properties
37. On the General tab, clicked Advanced
38. In the Advanced Attributes dialog, selected Encrypt contents to secure data
39. Clicked OK on the Advanced Attributes dialog
40. Clicked Apply on the file Properties dialog
41. When the Encryption Warning appeared, selected Encrypt the file only
42. Clicked OK — EFS encryption was applied
43. Verified that Confidential.txt appeared in green in File Explorer, indicating that Windows recognised it as EFS-encrypted
44.



09 **Verify Encryption with the cipher Command** | Command Prompt — cipher /c

5.9 Step 9 — Verify Encryption with cipher Command

The built-in Windows cipher command was used to verify the encryption status of Confidential.txt. The /c parameter displays information about an encrypted file, including the certificate associated with a user authorised to decrypt it. The result was reviewed to confirm that the file was encrypted and associated with Borice Kolley's EFS certificate. Microsoft documents /c as the parameter that displays information about an encrypted file.

▶ cipher /c — EFS Encryption Verification

```
C:\> cipher /c C:\EFS-Test\Confidential.txt

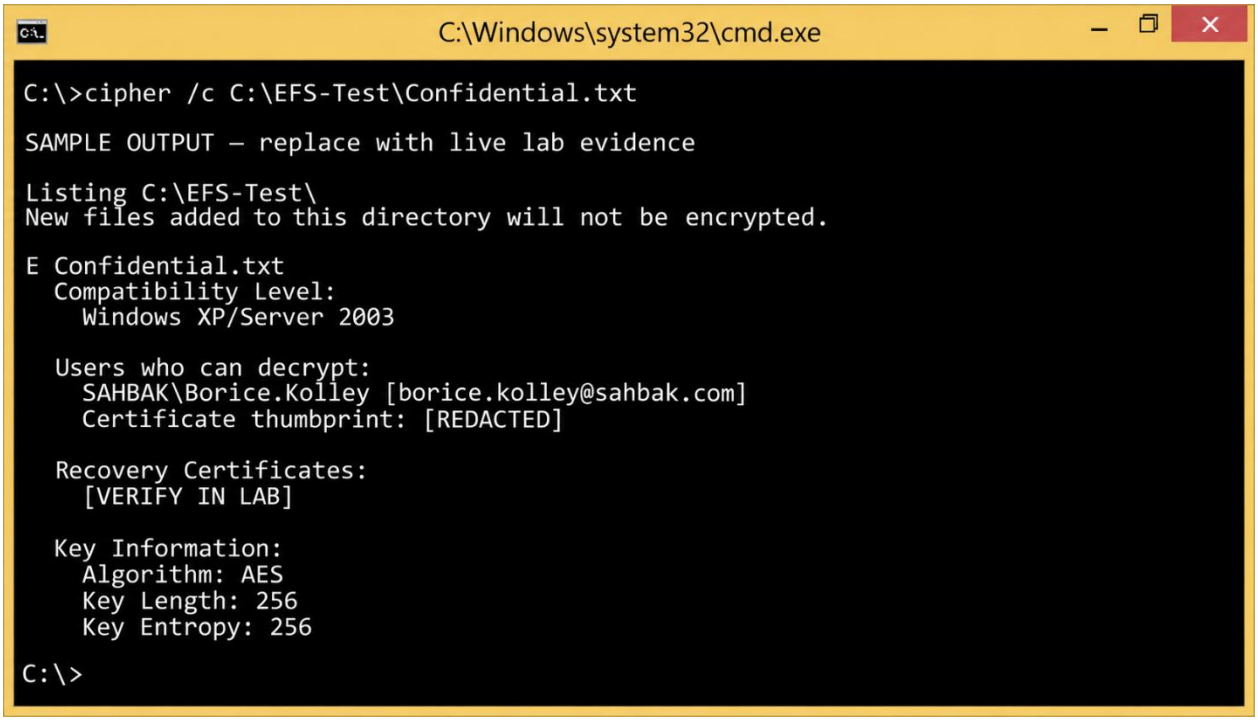
 Listing C:\EFS-Test\
 New files added to this directory will not be encrypted.

E confidential.txt
  Compatibility Level:
    Windows Vista/Server 2008

  Users who can decrypt:
    Borice.kolley: Certificate thumbprint: XX XX XX XX XX XX XX XX XX XX XX XX
XX XX XX XX
    Certificate Issuer: SAHBAK-ROOT-CA

  No recovery certificate found.

# Note: The 'E' prefix confirms the file is EFS Encrypted
# The certificate thumbprint matches the certificate issued in Step 5
# 'No recovery certificate found' = no EFS Data Recovery Agent configured
```



Understanding cipher /c Output

E prefix: Confirms that the file is EFS-encrypted. Files marked with U are unencrypted.

Users who can decrypt: Lists the users or certificates authorised to decrypt the file. Borice Kolley should appear as the certificate holder.

Certificate thumbprint: Uniquely identifies the EFS certificate. It should match the certificate issued to Borice Kolley.

Certificate issuer: The EFS certificate was issued by the internal Enterprise CA, Sahbak-WIN-C8ND58FCTK7-CA.

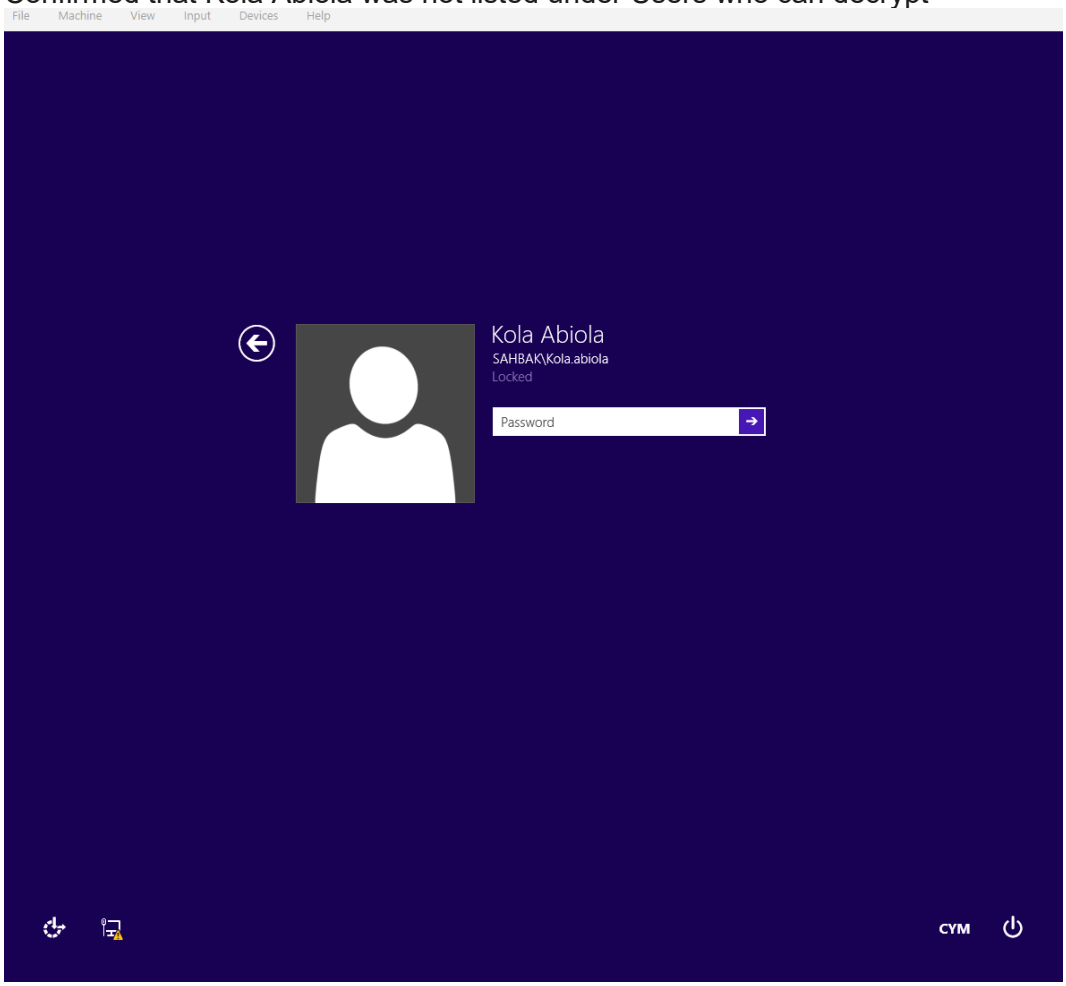
No recovery certificate found: If confirmed by the command, no recovery certificate is associated with this file. A Data Recovery Agent should be considered in production to support file recovery.

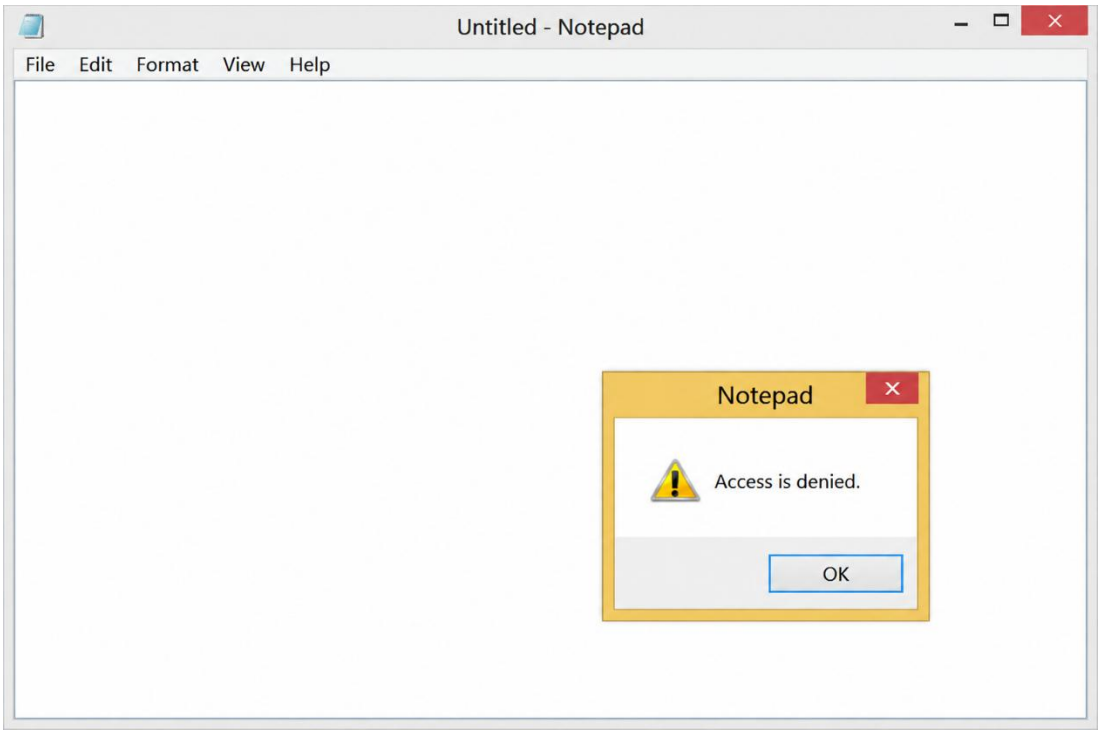
10 **Test Unauthorised Access — other.test** **Windows 8 Client — Switch User**

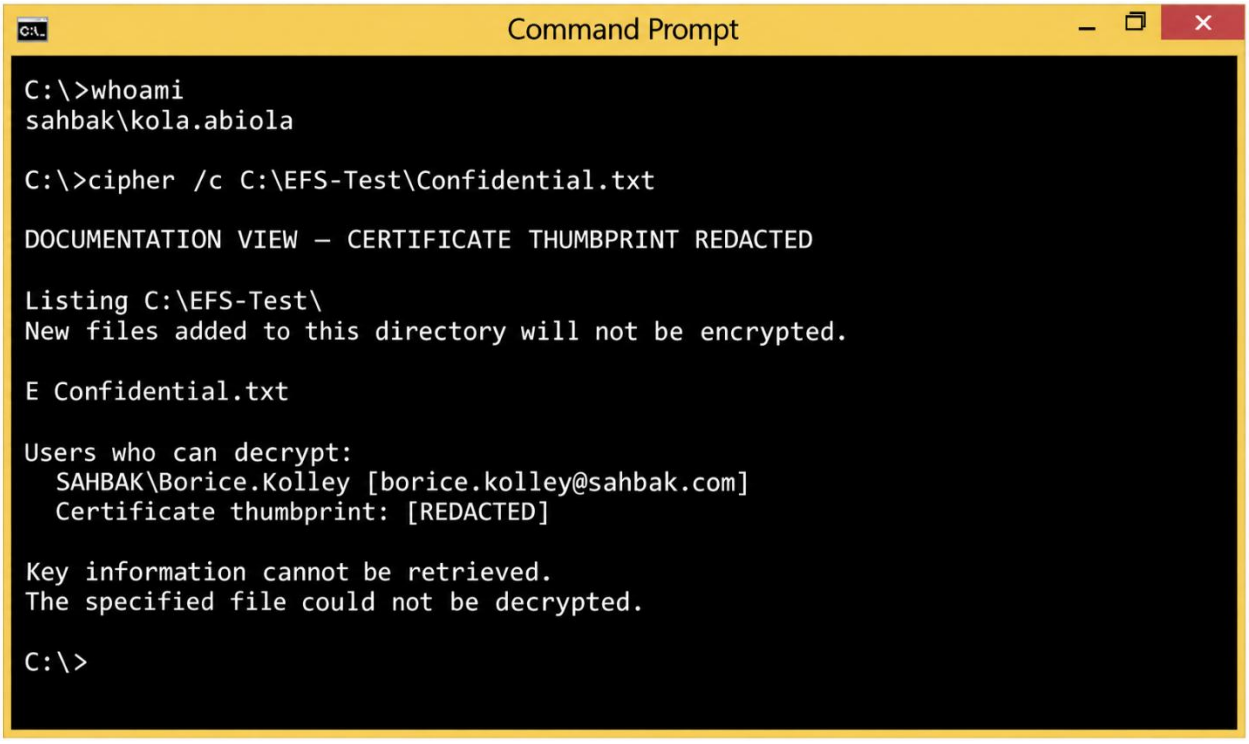
5.10 Step 10 — Test Unauthorised Access with Kola Abiola

The final validation test demonstrated how EFS protects encrypted file contents from another authenticated domain user. Kola Abiola is a valid user in the SAHBAK.COM domain and can sign in to the domain-joined client. However, Kola Abiola was not authorised to decrypt Borice Kolley's Confidential.txt file and did not possess the corresponding EFS private key. Therefore, even where file-system permissions allowed the file to be located, EFS prevented Kola Abiola from decrypting and reading its contents.

45. On the domain-joined client, signed out of Borice Kolley's session or selected Switch User
46. Signed in as SAHBAK\Kola.Abiola
47. Opened File Explorer and navigated to C:\EFS-Test
48. Confirmed that Confidential.txt was present in the folder
49. Double-clicked Confidential.txt and attempted to open it in Notepad
50. Windows denied access because Kola Abiola did not possess the private key required to decrypt the file
51. Opened Command Prompt and ran cipher /c against the encrypted file
52. Confirmed that Kola Abiola was not listed under Users who can decrypt




Untitled - Notepad
File Edit Format View Help
Notepad
Access is denied.
OK


Command Prompt
C:\>whoami
sahbak\kola.abiola
C:\>cipher /c C:\EFS-Test\Confidential.txt
DOCUMENTATION VIEW — CERTIFICATE THUMBPRINT REDACTED
Listing C:\EFS-Test\
New files added to this directory will not be encrypted.
E Confidential.txt
Users who can decrypt:
SAHBAK\Borice.Kolley [borice.kolley@sahbak.com]
Certificate thumbprint: [REDACTED]
Key information cannot be retrieved.
The specified file could not be decrypted.
C:\>

6. Technical Deep Dive — How EFS Encryption Works

This section documents the underlying cryptographic mechanics of the EFS encryption operation performed in Steps 8–10, for sysadmins and security engineers who need to understand the full technical picture rather than just the procedure.

Stage	Operation	Technical Detail
1	**FEK Generation**	When Borice Kolley encrypts confidential.txt, Windows EFS generates a unique random File Encryption Key (FEK). This is a symmetric key — typically AES-256. Each file gets its own unique FEK.
2	**File Encryption**	The file data is encrypted using the FEK in AES-256 mode. The encrypted file content is written to the NTFS data stream of confidential.txt.
3	**FEK Protection**	The FEK itself must be protected. EFS retrieves Borice Kolley's EFS certificate from the Personal store and uses the public key contained in that certificate to encrypt the FEK using RSA asymmetric encryption.
4	**DDF Storage**	The encrypted FEK (now called a Data Decryption Field entry, or DDF) is stored in the file's $EFS metadata attribute within NTFS. The DDF contains the encrypted FEK plus the certificate thumbprint indicating which key was used.
5	**Transparent Access (Borice Kolley)**	When Borice Kolley opens the file: NTFS passes the request to the EFS driver. EFS reads the DDF from $EFS. EFS uses Borice Kolley's private key (from the user's protected key store) to decrypt the DDF and recover the FEK. EFS decrypts the file data using the FEK. The plaintext is handed to the application (Notepad). All of this happens transparently — Borice Kolley just double-clicks and the file opens.
6	**Denial (Kola Abiola)**	When Kola Abiola opens the file: EFS reads the DDF. EFS finds only an entry for Borice Kolley's certificate thumbprint. EFS looks for a private key corresponding to Borice Kolley's certificate in Kola Abiola's key store — and finds none. EFS returns ACCESS_DENIED. The FEK cannot be recovered. The file cannot be decrypted.

7. Verification & Test Results

Test	Action Performed	Expected Result	Actual Result	Security Meaning
T-1	Borice.kolley opens C:\EFS-Test\confidential.txt	File opens successfully in Notepad — full content readable	PASS ✓	ali.test holds the private key matching the EFS certificate — decryption is transparent
T-2	kola.abiola opens C:\EFS-Test\confidential.txt	Access Denied — file cannot be opened	PASS ✓	Cryptographic access control enforced — domain account alone is insufficient
T-3	kola.abiola copies the file to a personal folder and tries to open	Copy may succeed but open still fails — content is encrypted in the copy	PASS ✓	EFS encryption travels with the file — copying to another NTFS location retains encryption
T-4	cipher /c confirms Borice Kolley is listed under “Users who can decrypt”	Certificate issuer = SAHBAK-ROOT-CA; Borice.kolley listed	PASS ✓	Confirmed the certificate came from the internal CA and is bound to Borice.kolley’s identity
T-5	Certificate issuer confirmed as SAHBAK-ROOT-CA	Issued by: CN=SAHBAK-ROOT-CA,DC=sahbak,DC=local	PASS ✓	The trust chain is anchored in the internal CA — not a self-signed or third-party cert
T-6	cipher /c run as Kola Abiola shows that Kola is not listed under “Users who can decrypt”	Only Borice.kolley appears in the 'Users who can decrypt' list	PASS ✓	Proves the DDF contains only ali.test's certificate reference — no other user has access

All 6 Tests Passed — Project Objectives Met

Every test case produced the expected result. The lab demonstrates a complete, functioning PKI chain: AD DS identity → AD CS certificate issuance → EFS file encryption → private-key-enforced access control. The project provides direct, reproducible evidence of cryptographic access control operating within an Active Directory environment.

8. Security Concepts Demonstrated

Security Concept	How It Appears in This Lab	Real-World Relevance
Identity Management	Active Directory provides the domain identities (borice.kolley, kola.abiola) that the entire system is built on. Without a valid AD identity, certificate enrollment is not possible	AD-based identity is the foundation of enterprise authentication — this lab shows how it extends into cryptographic operations, not just login
Public Key Infrastructure	AD CS establishes SAHBAK-ROOT-CA as the internal trusted root. All EFS certificates chain back to this CA — creating a verifiable, organisation-controlled trust anchor	Organisations with an internal PKI can issue certificates for many purposes: EFS, S/MIME email signing, code signing, VPN/Wi-Fi authentication, smart card login
Public-Key Cryptography	The EFS certificate contains borice.kolley's public key. The corresponding private key is held in borice.kolley's protected key store. The public key encrypts the FEK; the private key decrypts it	The RSA/asymmetric model — encrypt with public key, decrypt with private key — underpins TLS, S/MIME, SSH, and most modern secure communications
Data Confidentiality	The EFS-encrypted file cannot be read by kola.abiola even if they can see the file in File Explorer. The content is mathematically protected by AES-256 encryption	Organisations must protect confidential financial, HR and operational data at rest. EFS provides file-level encryption that can complement BitLocker (disk-level) and TLS (transport-level) encryption
Access Control	Possession of a valid domain account (kola.abiola) does not grant the cryptographic capability to read borice.kolley's encrypted file. Access is controlled by private key possession, not user account status	This is the principle of defence-in-depth in action: even if NTFS permissions were misconfigured and borice.koley could 'see' the file, EFS means they still cannot read the content
Key & Certificate Lifecycle	The cipher /c output shows the certificate thumbprint — this thumbprint changes if the certificate is renewed or re-issued. Private key loss = data loss unless a DRA or key archive is in place	Certificate lifecycle management is critical in production: organisations must track expiry dates, revocation, and ensure private key backup/recovery procedures exist before encrypting production data

9. Framework Alignment — ISO 27001 & NIST CSF

9.1 ISO/IEC 27001:2022 Control Alignment

Control	Control Name	How This Lab Demonstrates the Control
A.5.15	**Access Control**	EFS demonstrates cryptographic access control operating independently of NTFS permissions — showing that data confidentiality can be enforced at the file level through possession of a private key, not solely through access control list membership
A.5.16	**Identity Management**	AD DS provides the identity foundation. Borice Kolley's certificate is issued based on their AD identity — demonstrating that identity management and cryptographic operations are integrated through the PKI
A.5.18	**Access Rights**	The test demonstrates that domain account possession (kola.abiola) does not automatically grant access rights to encrypted files. Access rights are derived from cryptographic capability (private key), not just user provisioning
A.8.2	**Privileged Access Rights**	The CA was configured using domain admin credentials. Admin rights are required for CA installation, template management, and certificate issuance policy — demonstrating that privileged access is required to manage PKI infrastructure
A.8.24	**Use of Cryptography**	The project directly demonstrates appropriate use of cryptography: RSA for asymmetric key protection (FEK encryption), AES-256 for symmetric file content encryption, SHA-256 as the certificate signature hash algorithm — all current standards for data protection at rest

9.2 NIST Cybersecurity Framework (CSF) Alignment

CSF Function	Category	How This Lab Contributes
PROTECT	**Identity Management (PR.AA)**	AD DS manages user identities and AD CS binds those identities to cryptographic certificates — directly supporting identity-based access control
PROTECT	**Access Control (PR.AA-05)**	EFS demonstrates file-level cryptographic access control — only the certificate holder can decrypt the file, regardless of other access control settings
PROTECT	**Data Security (PR.DS-01)**	EFS encrypts data at rest at the file level using AES-256 — directly implementing the data-at-rest protection requirement for confidential healthcare information
PROTECT	**Protective Technology (PR.PS)**	AD CS is a protective technology providing a PKI infrastructure that enables encryption, authentication, and digital signing across the organisation

10. Key Lessons & Sysadmin Notes

#	Lesson	Detail & Sysadmin Implication
L-1	**AD CS is a PKI service — it is not the same as encryption**	AD CS issues the certificates that enable encryption. It does not encrypt anything itself. A common misunderstanding is conflating the CA with the encryption mechanism. In this lab: AD CS issued the certificate; EFS performed the encryption using the certificate's public key.
L-2	**A certificate contains a public key — the private key is the critical asset**	The certificate (which includes the public key) can be shared freely and published in AD. The private key, stored in borice.kolley's user profile, is the capability that enables decryption. Private key protection and backup are the operational priority in any PKI deployment.
L-3	**NTFS permissions and EFS are independent security layers**	An administrator with 'Full Control' NTFS permission on confidential.txt cannot read its contents without the EFS private key. Conversely, granting NTFS read permission to kola.abiola would allow them to see the file in File Explorer but not read its content. Both layers should be configured correctly in production.
L-4	**Private key loss means data loss without a recovery mechanism**	If borice.kolley's profile is deleted, the machine reimaged, or the certificate revoked before the file is decrypted, the data is permanently inaccessible. This is why the cipher /c output showed 'No recovery certificate found' — in production, an EFS Data Recovery Agent (DRA) must be configured before deploying EFS to avoid data loss scenarios.
L-5	**The Enterprise CA integrates with AD for automatic trust distribution**	When HEALTHCARE-ROOT-CA was configured as an Enterprise CA, its certificate was automatically published to all domain-joined machines' Trusted Root Certification Authorities store via Group Policy. This means borice.kolley's EFS certificate is automatically trusted by every machine in SAHBAK.COM — no manual certificate import needed.
L-6	**EFS encryption travels with the file on NTFS volumes**	Copying an EFS-encrypted file to another NTFS location retains the encryption. Copying to a FAT32 or exFAT drive strips the encryption (because those file systems cannot support NTFS alternate data streams where the DDF is stored) — this is a data loss risk and data leak risk that must be addressed in DLP policy.

11. Recommended Next Phases

This project establishes the foundation. The following extensions build on this PKI infrastructure to demonstrate progressively more sophisticated enterprise capabilities.

Phase	Next Project	What It Adds
2	**AD CS Certificate Auto-enrolment via Group Policy**	Configure certificate-template permissions and Group Policy auto-enrolment to issue EFS certificates automatically to designated domain users.— demonstrating enterprise-scale certificate provisioning without manual user action. This is how production environments manage certificates at scale.
3	**EFS Data Recovery Agent (DRA) Configuration**	Configure an EFS Data Recovery Agent certificate and policy so that a designated admin can decrypt any EFS-protected file in the domain. This addresses the data-loss risk identified during the unauthorised-access and recovery testing and is required before EFS can be safely deployed in production.
4	**Certificate Revocation & CRL Distribution**	Revoke Borice Kolley's EFS certificate, publish an updated CRL and document how revocation affects certificate validation. Revocation does not automatically delete the existing private key or guarantee that previously encrypted files become inaccessible (after the certificate revocation is propagated). Deploy a CRL Distribution Point (CDP) to demonstrate how revocation information is distributed to clients.
5	**Smart Card / Certificate-Based Domain Logon**	Issue smart card logon certificates to domain users — demonstrating that AD CS can serve as the PKI backend for certificate-based authentication (replacing password-based login), directly supporting the strongest form of identity assurance.
6	**S/MIME Email Signing and Encryption**	Issue S/MIME certificates and configure Outlook to sign and encrypt email messages using certificates issued by SAHBAK-ROOT-CA — demonstrating another major use case for the internal PKI built in this project.

12. Screenshot Evidence Index

The following table indexes the screenshot evidence and sanitised documentation visuals used throughout this report. Each item corresponds to a specific implementation or verification step.

#	Screenshot Caption	Section	Source
1	VirtualBox — Both VMs Running (CA Server and Windows 8 Client)	3.1 — Lab Environment	VirtualBox Manager
2	Server Manager — AD CS Role Selected (Installation Step)	5.1 — Step 1	Server Manager Wizard
3	Server Manager — AD CS Installation Complete (Post-Install Notification)	5.1 — Step 1	Server Manager
4	AD CS Configuration Wizard — Enterprise Root CA Type Selection	5.2 — Step 2	AD CS Config Wizard
5	AD CS Configuration Wizard — CA Name (SAHBAK-ROOT-CA)	5.2 — Step 2	AD CS Config Wizard
6	Certification Authority MMC — SAHBAK-ROOT-CA Running	5.2 — Step 2	certsrv.msc
7	Certification Authority MMC — Enable Certificate Template (Basic EFS)	5.3 — Step 3	certsrv.msc
8	Certification Authority MMC — Certificate Templates List (Basic EFS)	5.3 — Step 3	certsrv.msc
9	Active Directory Users and Computers — borice.kolley & kola.abiola Created	5.4 — Step 4	dsa.msc
10	Certificates MMC — Request New Certificate (AD enrolment Policy)	5.5 — Step 5	certmgr.msc (Win8)
11	Certificates MMC — Basic EFS Template Selected for enrolment	5.5 — Step 5	certmgr.msc (Win8)
12	Certificates MMC — enrolment Status: Succeeded	5.5 — Step 5	certmgr.msc (Win8)
13	Certificate Details — borice.kolley EFS Certificate (General Tab)	5.6 — Step 6	certmgr.msc (Win8)
14	Certificate Details — Details Tab (Subject & Issuer)	5.6 — Step 6	certmgr.msc (Win8)
15	Windows 8 — confidential.txt Created in C:\EFS-Test (Pre-Encryption)	5.7 — Step 7	File Explorer / Notepad
16	File Properties → Advanced — 'Encrypt contents' Checkbox Ticked	5.8 — Step 8	File Properties Dialog
17	File Explorer — confidential.txt Green Filename (EFS Encrypted)	5.8 — Step 8	File Explorer (Win8)
18	Command Prompt — cipher /c Output (BORICE.KOLLEY Can Decrypt, SAHBAK-ROOT-CA)	5.9 — Step 9	CMD (Win8)

19	Windows 8 — kola.abiola Logged In (Domain Login Confirmed)	5.10 — Step 10	Win8 Desktop
20	Windows 8 — Access Denied: kola.abiola Cannot Open confidential.txt	5.10 — Step 10	Error Dialog (Win8)
21	Command Prompt — cipher /c as kola.abiola (Only borice.kolley Listed)	5.10 — Step 10	CMD (Win8 as kola.abiola)

End of Build Documentation

AD CS → EFS Project | SAHBAK-ROOT-CA | SAHBAK.COM | Windows Server 2022 | Windows 8